

Pharmacology Lab 5

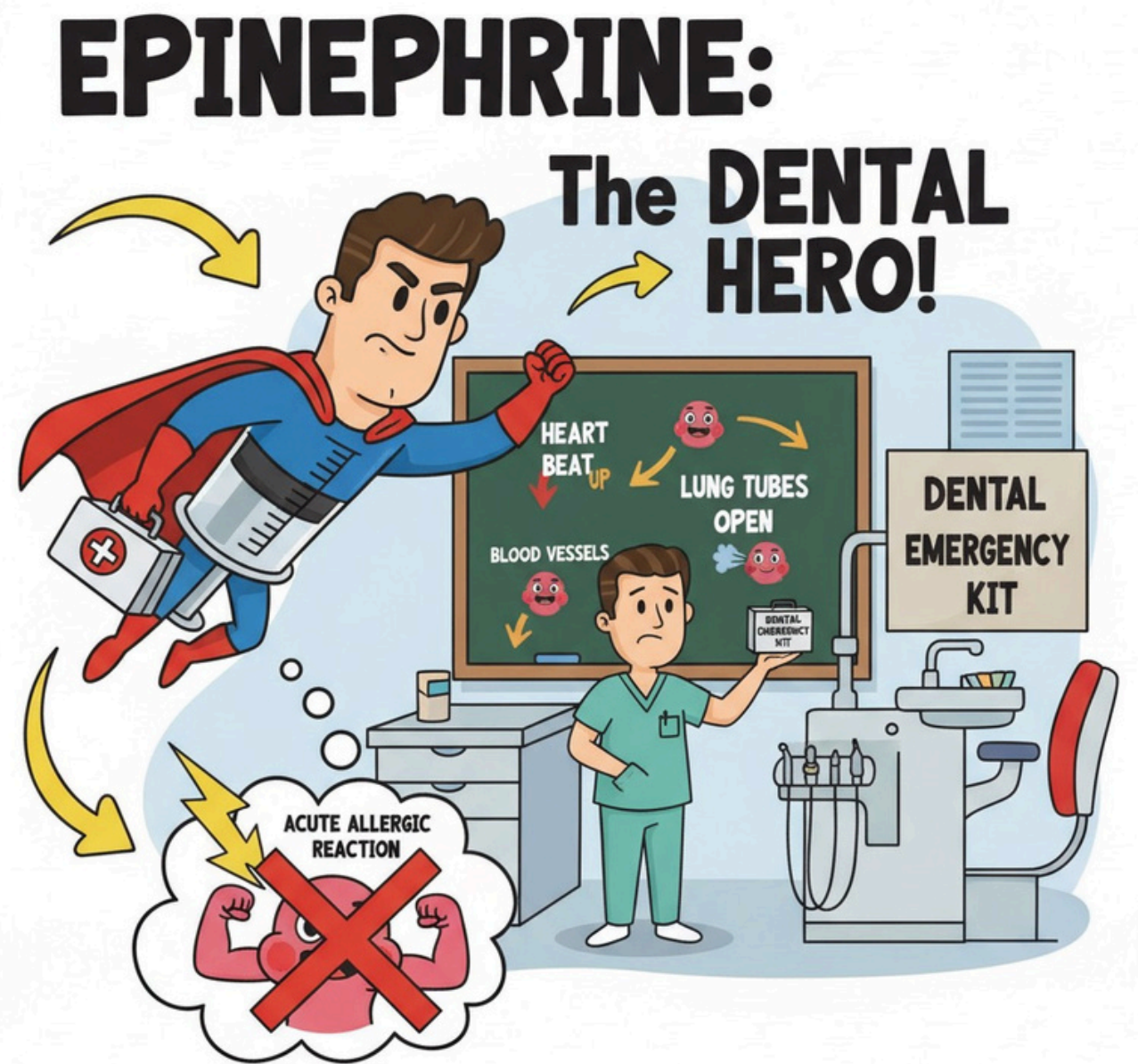
Epinephrine



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Epinephrine

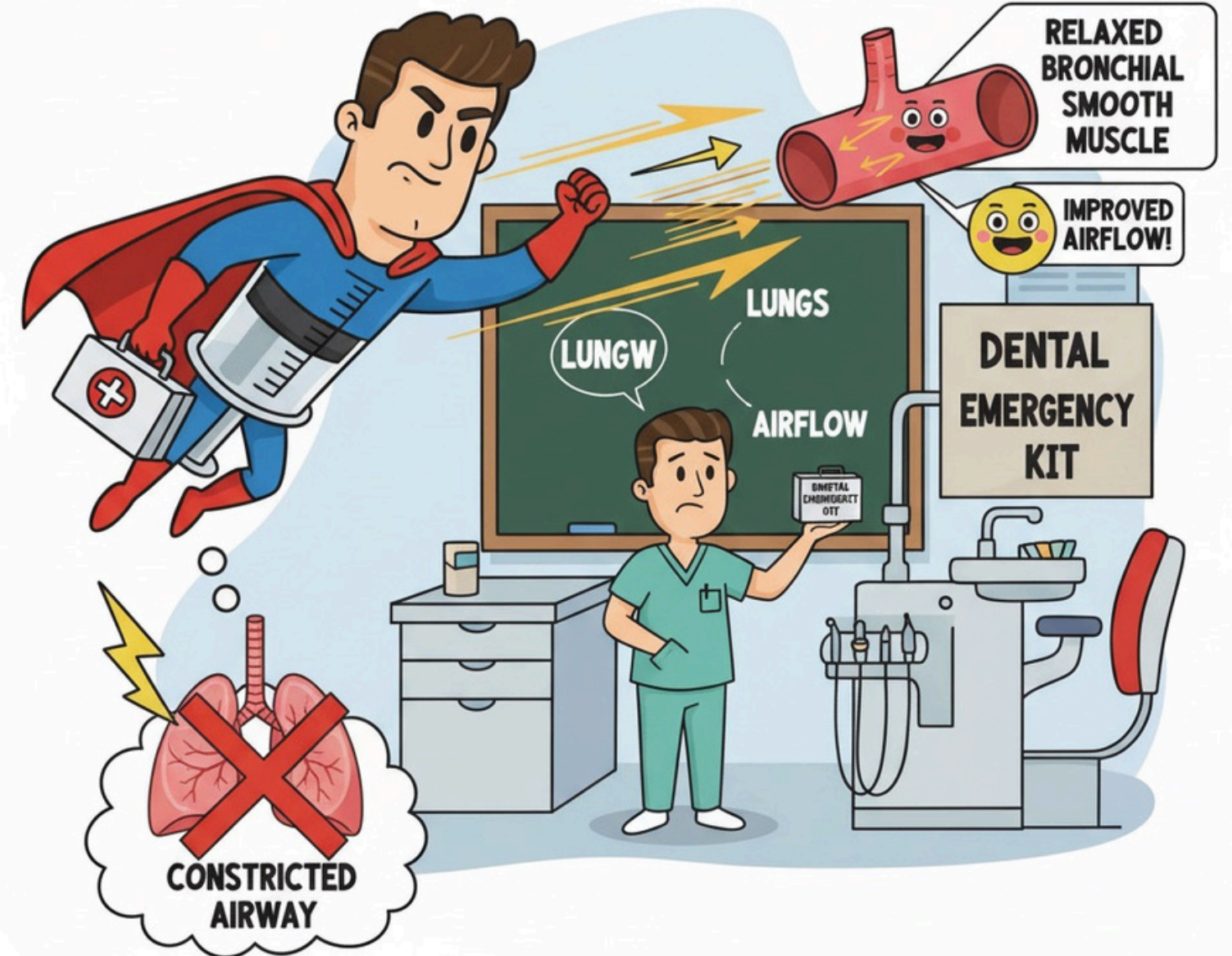
Epinephrine is a critical component of dental emergency kits, essential for managing acute allergic reactions and other medical emergencies. As an endogenous catecholamine, it stimulates both α - and β -adrenergic receptors, leading to several beneficial pharmacologic effects:



Bronchodilation:

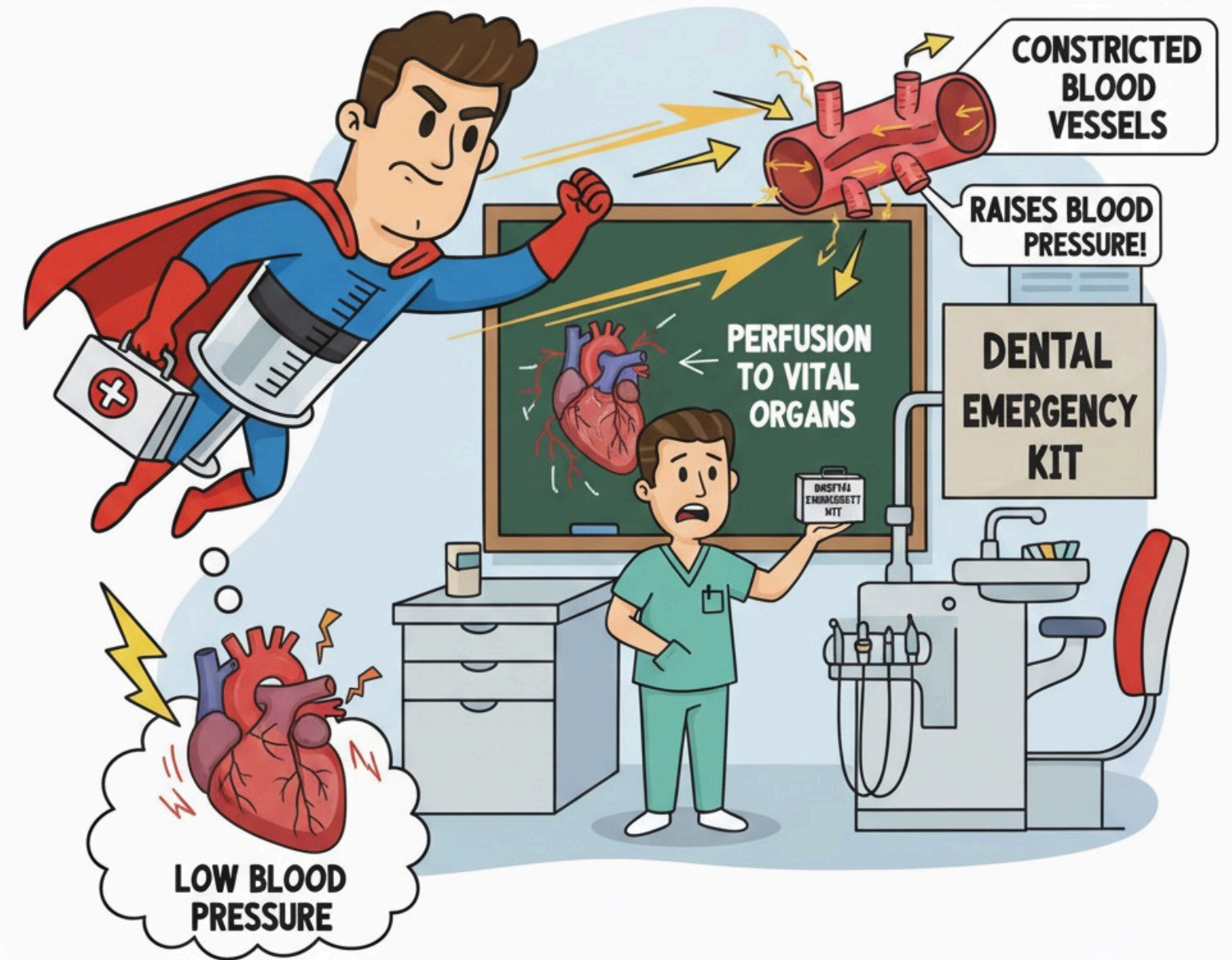
Relaxes bronchial smooth muscles, improving airflow and alleviating respiratory distress.

EPINEPRINE: Bronchodilation!



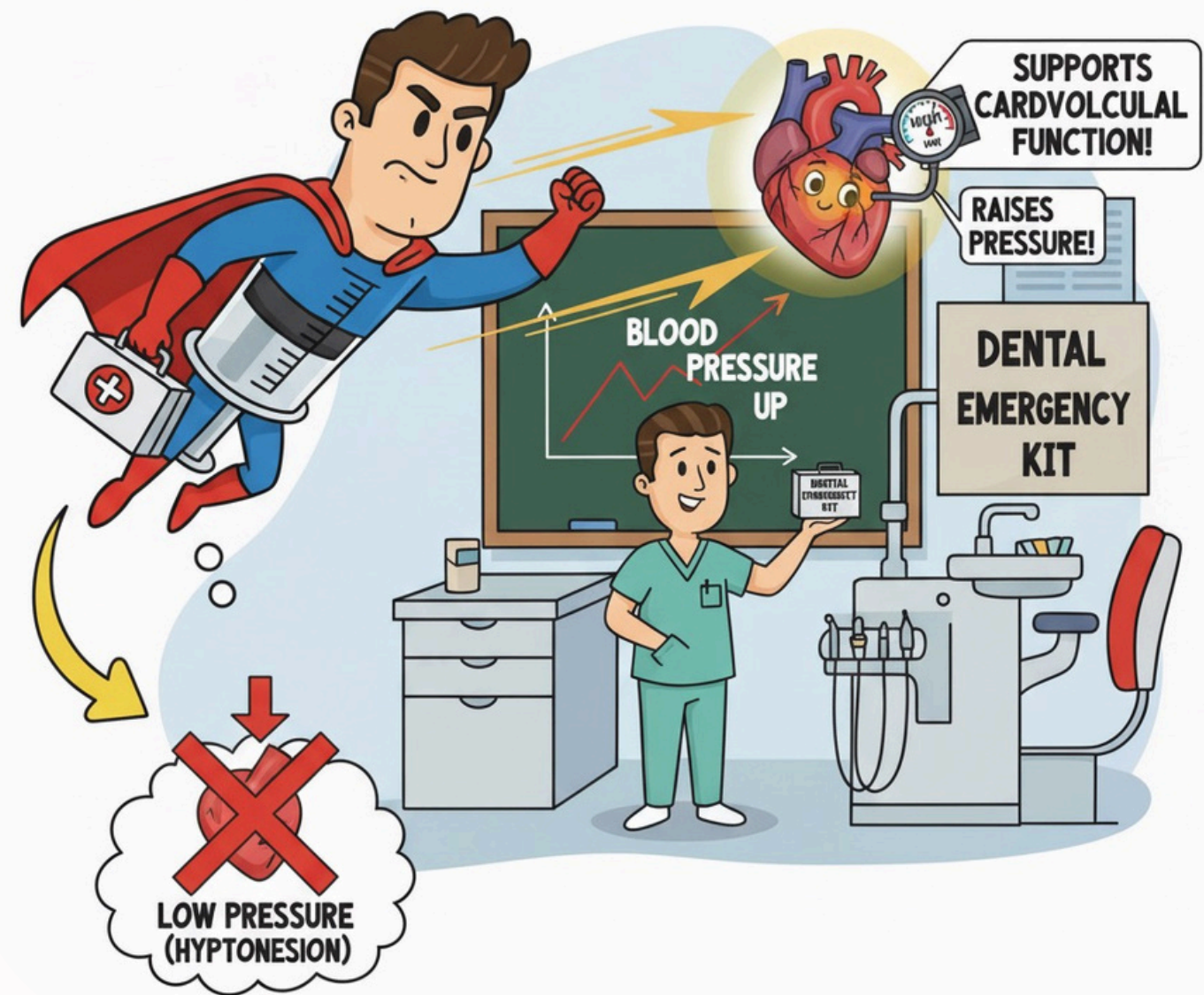
Increased Systemic Vascular Resistance:
Constricts blood vessels, raising blood pressure and enhancing perfusion to vital organs.

EPINEPRINE: Increased Systemic Vascular Resistance!



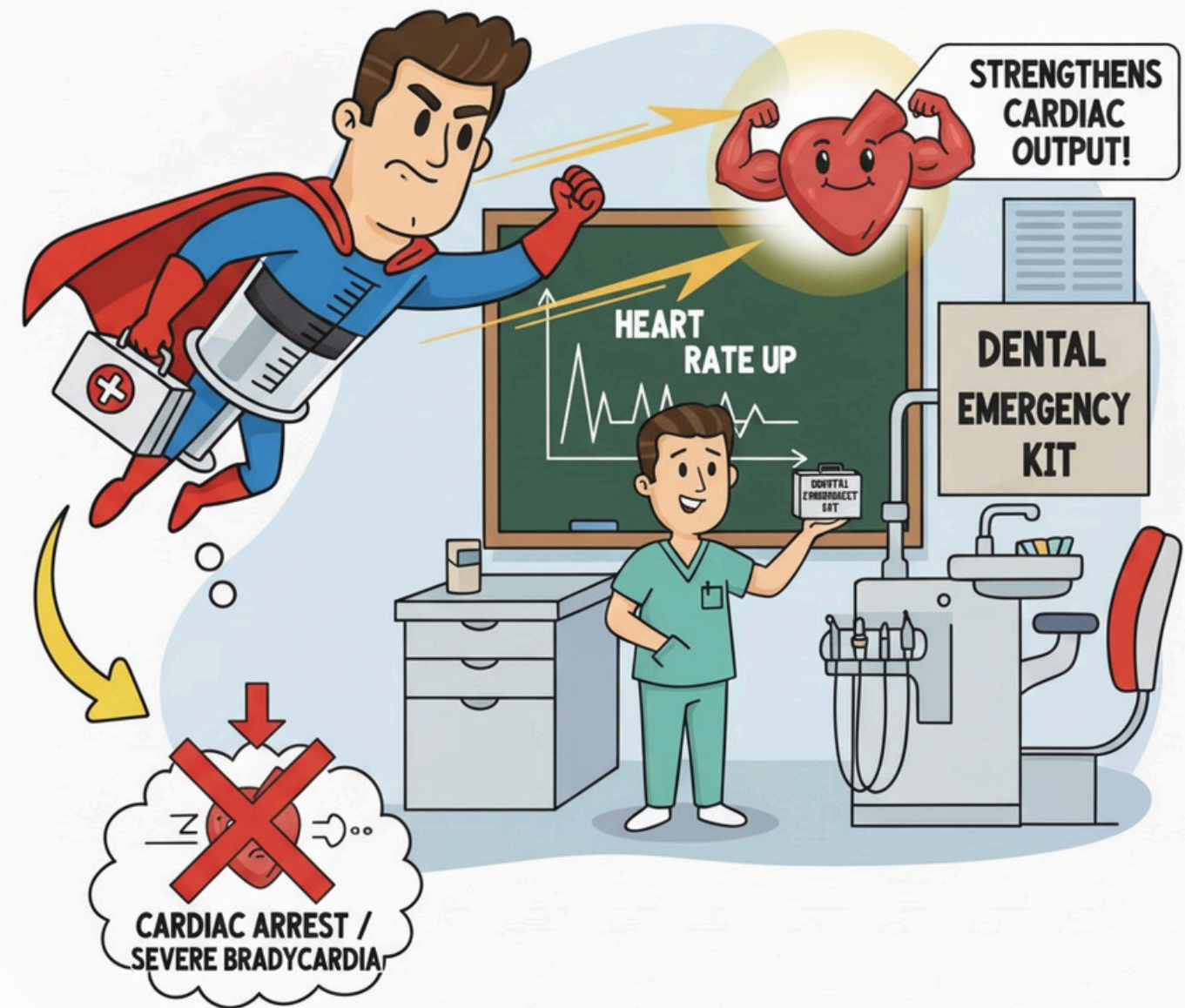
Elevated Arterial Blood Pressure:
Supports cardiovascular function
during hypotensive states.

EPINEPRINE: Elevated Elevated Arterial Blood Pressure!



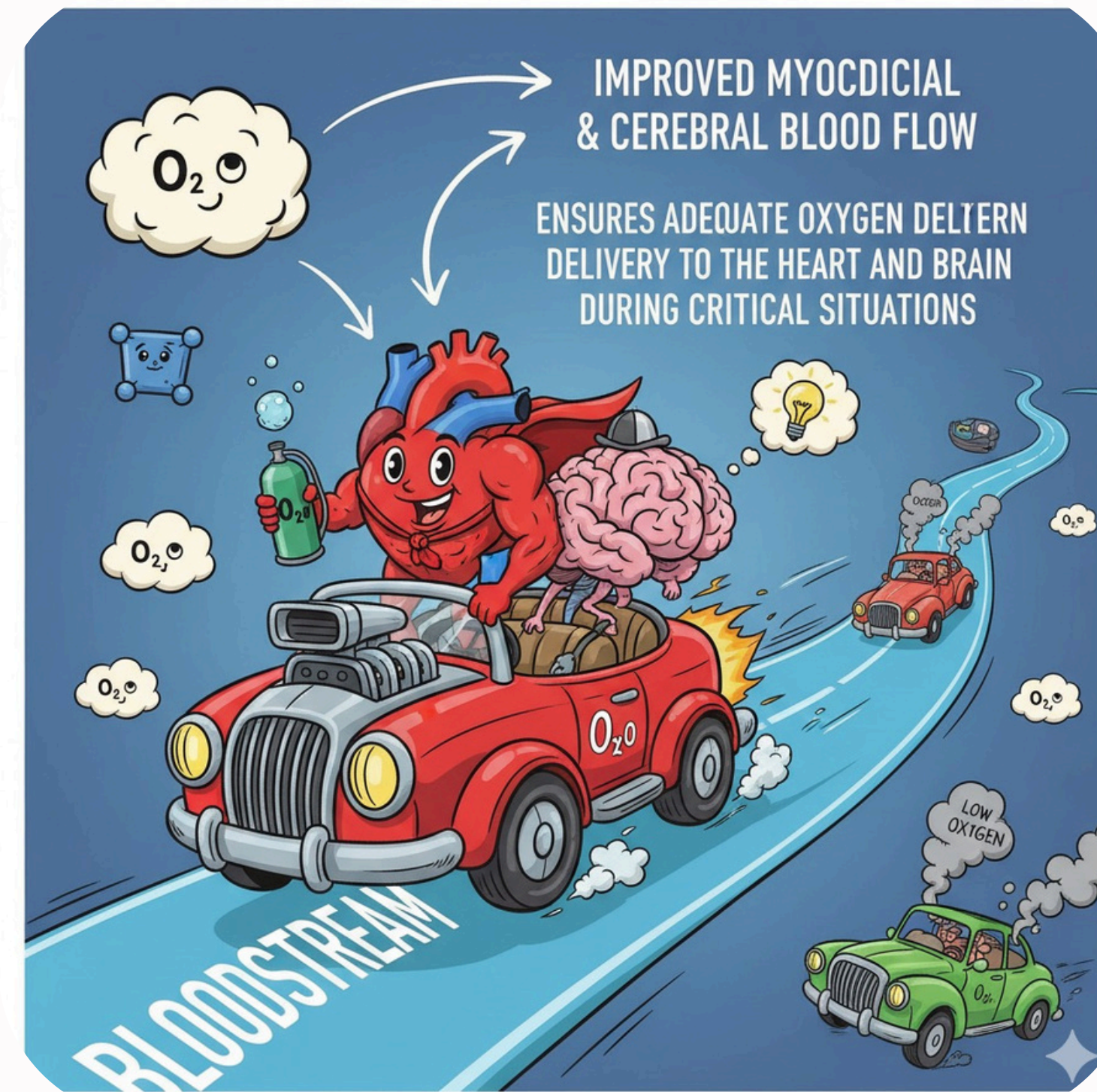
Enhanced Heart Rate and Myocardial Contractility :
Strengthens cardiac output, crucial during cardiac arrest or severe bradycardia.

EPINEPRINE: Enhanced Heart Rate and Myatocial Contractility!



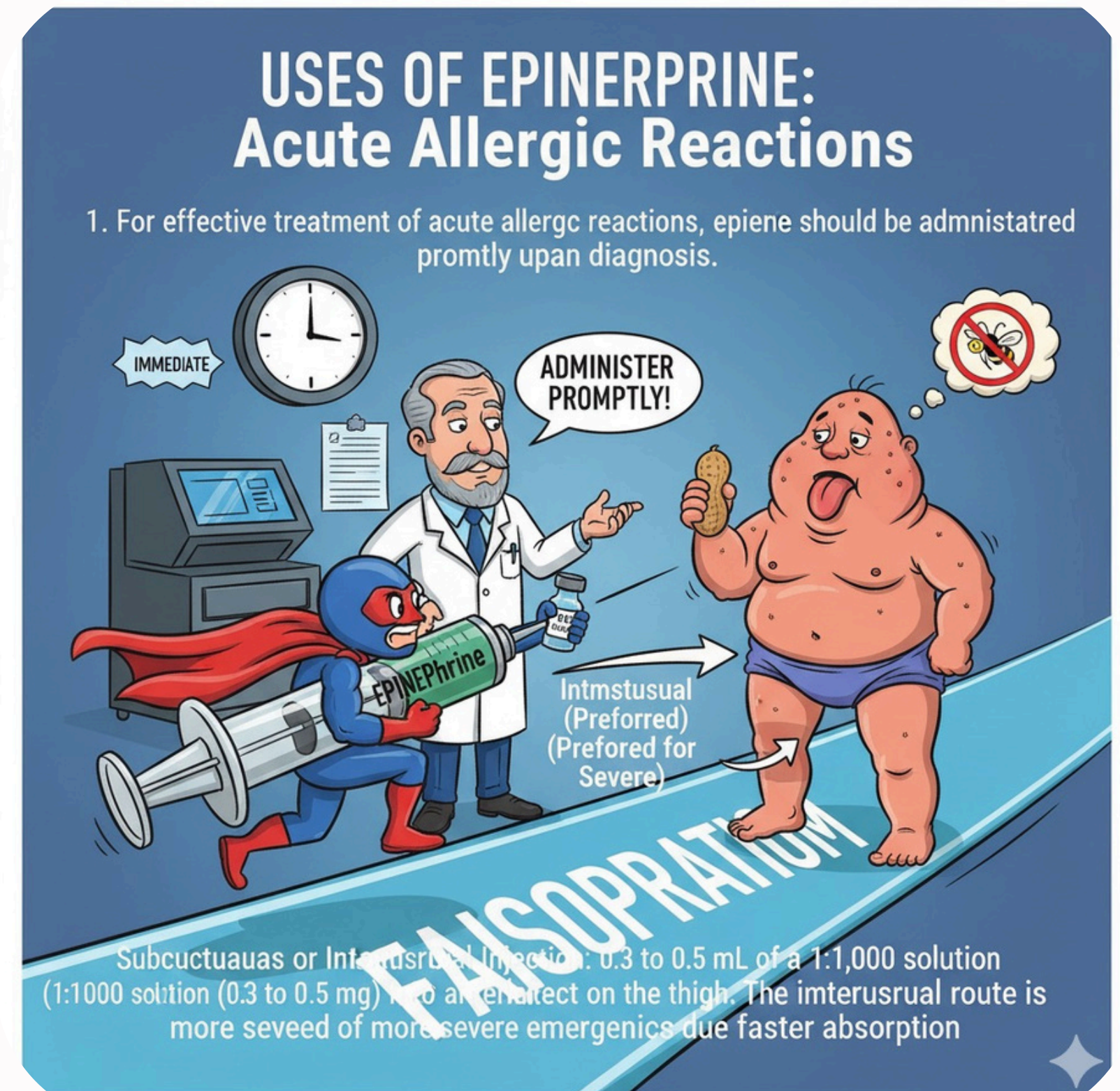
Improved Myocardial and Cerebral Blood Flow :

Ensures adequate oxygen delivery to the heart and brain during critical situations.



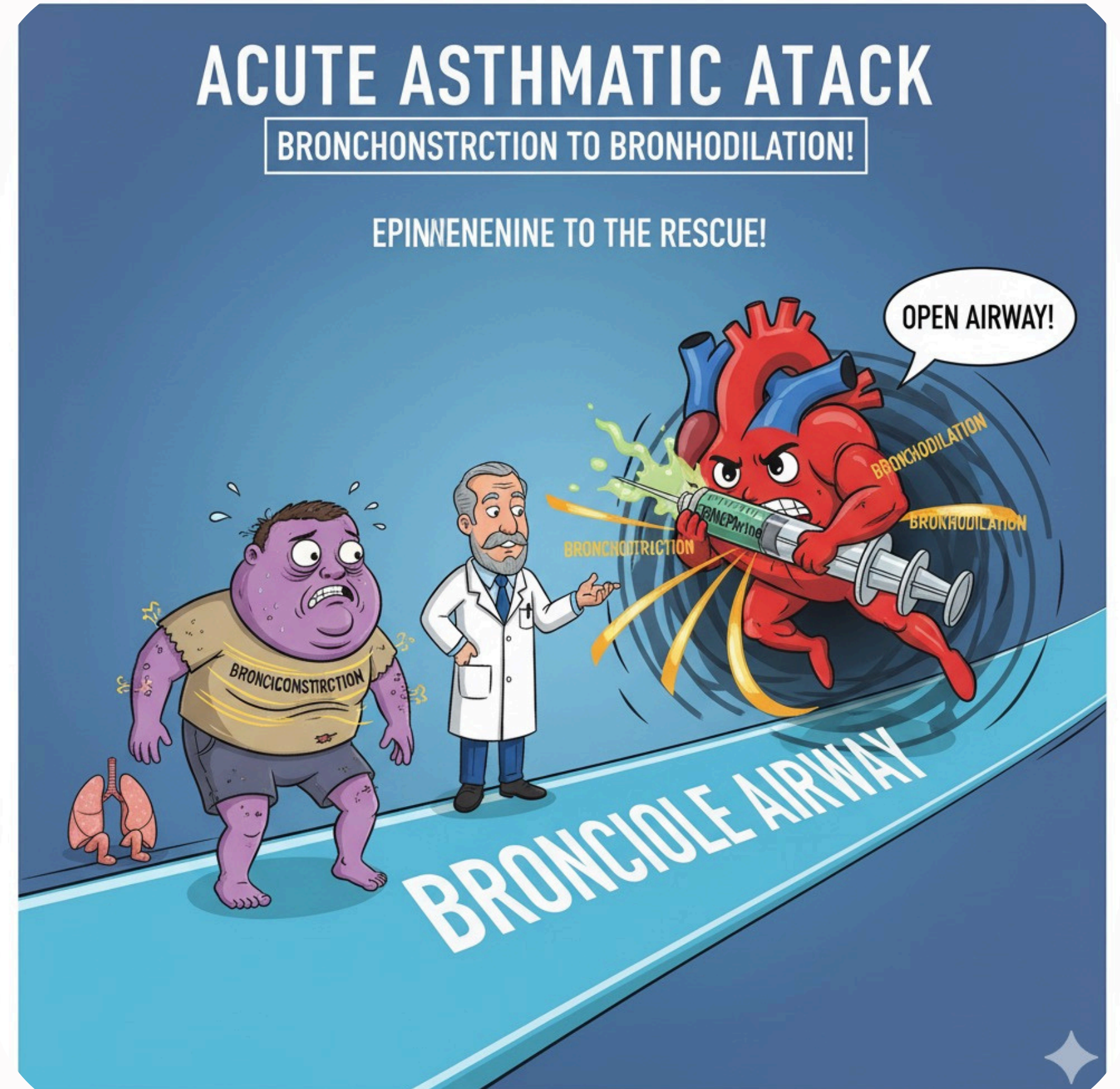
Uses of Epinephrine

1. For effective treatment of acute allergic reactions, epinephrine should be administered promptly upon diagnosis. The recommended dosages are: Subcutaneous or Intramuscular Injection : 0.3 to 0.5 mL of a 1:1,000 solution (0.3 to 0.5 mg) into the anterolateral aspect of the thigh. The intramuscular route is preferred for more severe emergencies due to faster absorption.



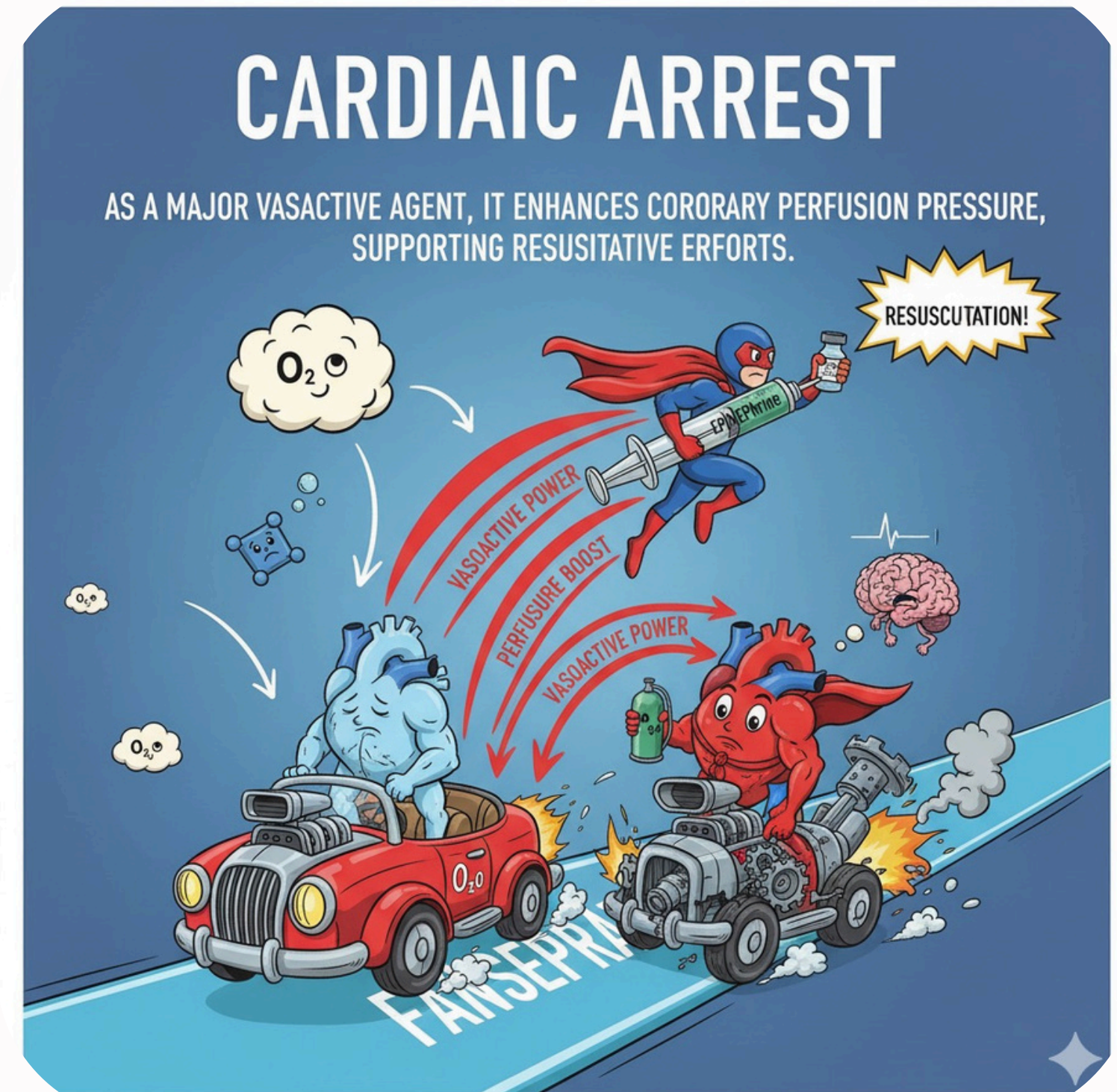
Acute Asthmatic Attacks:

Particularly when unresponsive to inhaled β 2-adrenergic agonists, due to its potent bronchodilating effects.



3. Injectable Solution of 0.1 mg/mL (1:10,000): 0.1 to 0.25 mg (1 to 2.5 mL) IV slowly once

4. Cardiac Arrest: As a major vasoactive agent, it enhances coronary perfusion pressure, supporting resuscitative efforts.



In dental settings, having epinephrine on hand is essential for rapidly addressing severe allergic reactions, such as to latex or certain medications.

DENTAL EMERGENCIES: ALLERGIC REACTIONS

EPIPINEHRINE: RAPID RESPONSE FOR SEVERE ALLRGIES



ESSENTIAL IN DENTAL SETTINGS FOR LATEX & MEDICATION ALLERGIES.

روي عن أمير المؤمنين ع
(من عَظَم صِغَارِ الْمَصَائِبِ إِبْتَلَاهُ اللَّهُ بِكِبَارِهَا)
صدق أمير المؤمنين ع

The image features a white background with three teal-colored decorative shapes in the corners: a quarter-circle in the top-left, a quarter-circle in the top-right, and a larger, more complex shape in the bottom-right. A central teal rounded rectangle contains the text "Thank you...." in white.

Thank you....